

CSE260 Assignment - 3

Deadline - 24th April

Marks : 40

Assignment must be handwritten. Submit the Hardcopy.

Make sure your circuit is efficient, that is using the lowest number of components. Use external gates if necessary.

Question 1: (5 Marks)

Design a half-adder using NAND gates only.

Question 2 (5 Marks)

Build a circuit that converts a 3-bit 2's complement number to its actual 3-bit binary number using encoder(s) and decoder(s).

Question 3: (5 Marks)

Design a 14-person voter counting system using full adders and parallel adders.

Question 4: (5 * 3 = 15 Marks)

Implement the following boolean function using

a) single 16:1 mux **b)** single 8:1 mux **c)** single 4:1 mux

$F(A,B,C,D) = \sum((0,1,2,4,5,7,11,12,14))$. Use external gates if required.

Question 5: (5 * 2 = 10 Marks)

Implement the following boolean function using

a) 4x16 decoder(s) only **b)** 2x4 decoder(s) only

$F(A,B,C,D,E) = \sum(0,1,5,15,24,25,27)$. Use external gates if required.

Ungraded Problems for Practice (not required to submit in Assignment-3):

Question 1

Design a BCD to Excess-6 code converter encoder(s) and decoder(s).

Question 2

Implement the following boolean function using both 4:1 and 2:1 mux in a single circuit.

$F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.

Question 3

Build a circuit that converts a 3-bit 1's complement number to its actual 3-bit binary number using encoder(s) and decoder(s).

Question 4

Design a half-adder using NOR gates only.

Question 5

Design a full adder using **a)** NOR gates only. **b)** NAND gates only

Question 6

Construct a 5x32 Decoder using only 3x8 Decoder(s)

Question 7

Construct the circuit of 4x2 Priority Encoder.

If the number of 1 in binary input is even, the priority is given to the MSB 1.

If the number of 1 in binary input is odd, the priority is given to the LSB 1.

Find the truth table first, then draw the KMAP, and finally construct the circuit of the outputs.